

Homework for Chap 16

For GaAs, the conduction band deformation potential $\mathcal{E}_c = 7.0 \text{ eV}$, and the valence band deformation potential $\mathcal{E}_v = -6.0 \text{ eV}$, and suppose the strain $\epsilon = \Delta a/a = 10^{-3}$:

1. Calculate the change of the band gap energy ΔE_g ;
2. Plot schematically how the conduction band and valence band shift compared with the unstrained GaA.